

Learned features for acquisition

Question: does a small learned vector from the molecular predictor improve a feedforward acquisition scorer beyond predicted mean and standard deviation?

Comparison: mean + SD alone, or mean + SD plus 2, 4, 8 or 16 learned features. Single-task predictors, K=1, five linked-group folds and three independently initialized PPO policies. Each predictor is an ensemble of three members.

The primary panel contains 580 held-out series with at least 15 compounds from 283 linked groups. All 4625 eligible series with at least 3 compounds are also evaluated. Counts are averaged within each endpoint, then across endpoints.

Primary top1 results: Mean + SD: 6.325; + 2 features: 6.551; + 4 features: 6.523; + 8 features: 6.733; + 16 features: 6.748.

None of the fixed extra-feature dimensions has a clearly positive paired improvement interval on the primary aggregate.

This isolates acquisition features. The frozen predictor gives the same mean and SD across arms at any identical measured state.

Mean acquisitions, series with at least 15 compounds

Top-1

	Mean+SD	+2	+4	+8	+16
Average	6.325	6.551	6.523	6.733	6.748
Microsomal CL	5.147	5.281	5.364	5.510	5.613
In vivo CL	9.458	9.054	8.412	9.025	8.871
Protein binding	4.845	5.208	5.233	5.627	5.674
Cellular CL	5.267	5.834	6.137	6.138	6.127
Permeability	7.933	7.910	7.947	8.019	8.081
Efflux ratio	5.299	6.022	6.043	6.079	6.123

Top-2

	Mean+SD	+2	+4	+8	+16
Average	4.255	4.386	4.492	4.462	4.528
Microsomal CL	3.766	3.677	3.663	3.652	3.732
In vivo CL	5.917	5.571	5.654	5.658	5.521
Protein binding	3.028	3.193	3.254	3.277	3.286
Cellular CL	3.438	3.887	4.245	4.169	4.303
Permeability	5.384	5.564	5.623	5.499	5.573
Efflux ratio	4.000	4.426	4.515	4.515	4.753

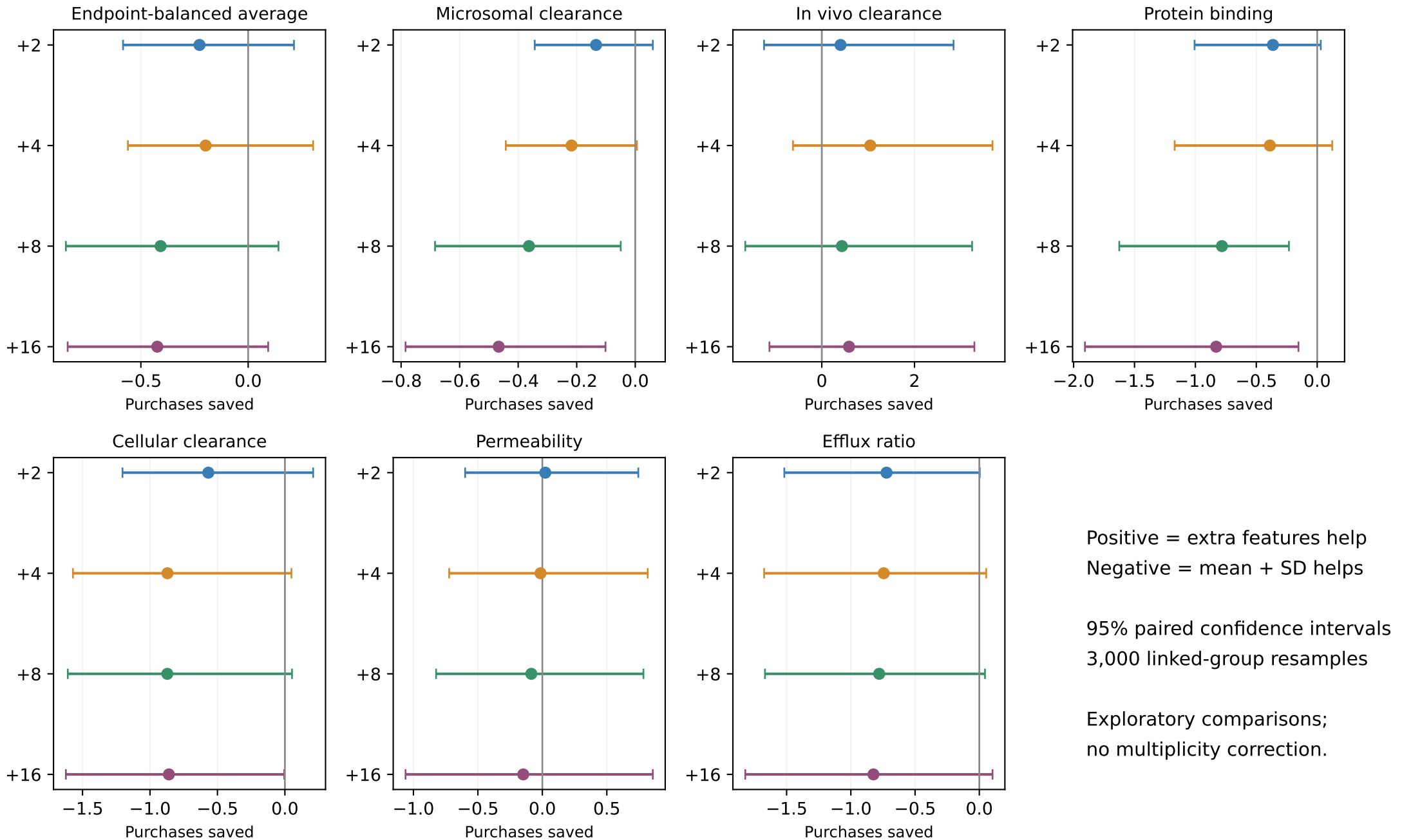
Top-3

	Mean+SD	+2	+4	+8	+16
Average	3.407	3.433	3.482	3.502	3.602
Microsomal CL	3.047	2.873	2.890	2.856	2.860
In vivo CL	4.383	4.054	3.896	4.050	4.267
Protein binding	2.303	2.263	2.261	2.316	2.360
Cellular CL	3.142	3.403	3.738	3.657	3.729
Permeability	4.263	4.337	4.425	4.260	4.341
Efflux ratio	3.302	3.668	3.681	3.875	4.057

Top-4

	Mean+SD	+2	+4	+8	+16
Average	2.768	2.846	2.892	2.890	2.941
Microsomal CL	2.655	2.522	2.531	2.504	2.515
In vivo CL	3.608	3.362	3.192	3.400	3.362
Protein binding	1.886	1.797	1.782	1.820	1.905
Cellular CL	2.546	2.843	3.211	3.077	3.161
Permeability	3.517	3.650	3.692	3.694	3.694
Efflux ratio	2.394	2.903	2.943	2.847	3.008

Top-1 improvement, series with at least 15 compounds



Mean acquisitions, series with at least 3 compounds

Top-1

	Mean+SD	+2	+4	+8	+16
Average	2.645	2.750	2.767	2.768	2.781
Microsomal CL	2.519	2.571	2.584	2.626	2.630
In vivo CL	2.523	2.617	2.591	2.602	2.587
Protein binding	2.054	2.121	2.120	2.161	2.172
Cellular CL	2.684	2.796	2.873	2.814	2.807
Permeability	3.229	3.320	3.342	3.308	3.363
Efflux ratio	2.863	3.075	3.094	3.098	3.127

Top-2

	Mean+SD	+2	+4	+8	+16
Average	1.831	1.891	1.912	1.896	1.908
Microsomal CL	1.830	1.839	1.836	1.838	1.856
In vivo CL	1.712	1.709	1.728	1.714	1.710
Protein binding	1.425	1.481	1.467	1.487	1.489
Cellular CL	1.839	1.950	2.000	1.976	1.990
Permeability	2.166	2.263	2.276	2.237	2.251
Efflux ratio	2.017	2.106	2.162	2.126	2.153

Top-3

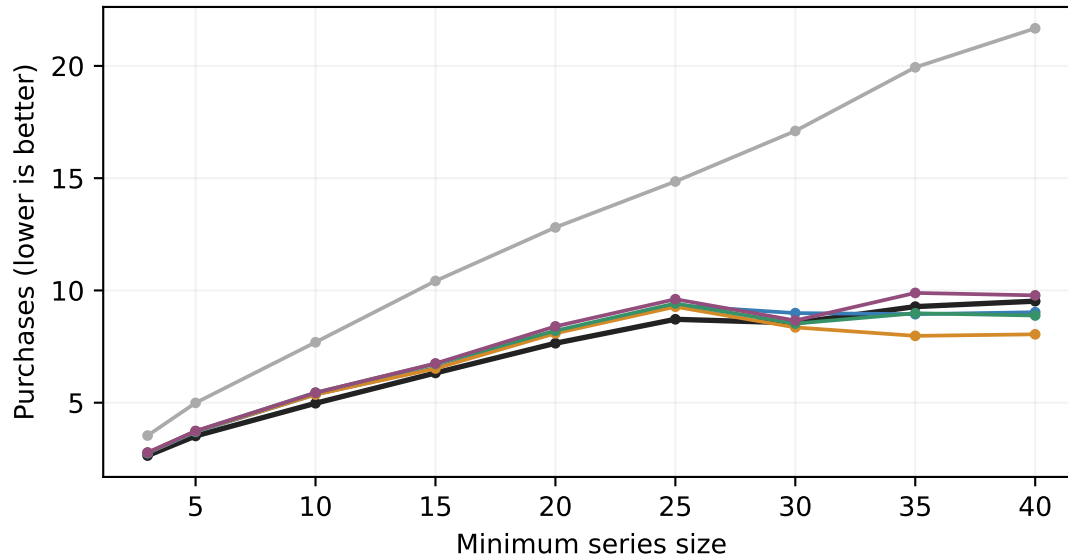
	Mean+SD	+2	+4	+8	+16
Average	1.152	1.198	1.216	1.208	1.216
Microsomal CL	1.180	1.167	1.173	1.170	1.174
In vivo CL	0.903	0.906	0.906	0.901	0.914
Protein binding	0.771	0.802	0.802	0.803	0.803
Cellular CL	1.213	1.299	1.346	1.320	1.324
Permeability	1.467	1.536	1.561	1.530	1.524
Efflux ratio	1.377	1.478	1.508	1.522	1.560

Top-4

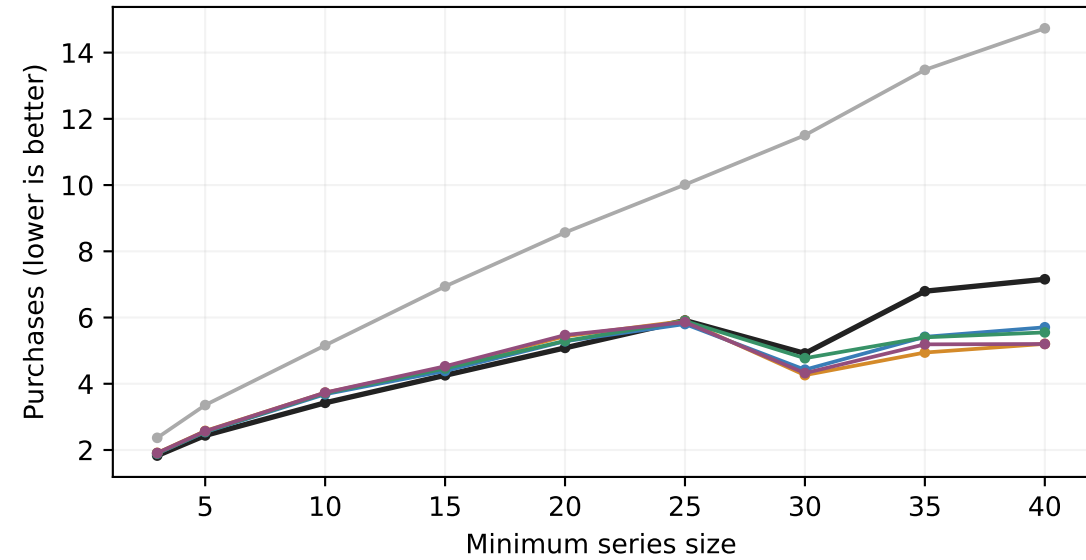
	Mean+SD	+2	+4	+8	+16
Average	0.807	0.849	0.860	0.848	0.858
Microsomal CL	0.889	0.882	0.878	0.877	0.881
In vivo CL	0.561	0.559	0.557	0.550	0.557
Protein binding	0.487	0.500	0.500	0.499	0.501
Cellular CL	0.854	0.940	0.991	0.956	0.970
Permeability	1.081	1.135	1.151	1.140	1.135
Efflux ratio	0.970	1.078	1.084	1.065	1.103

Endpoint-balanced average

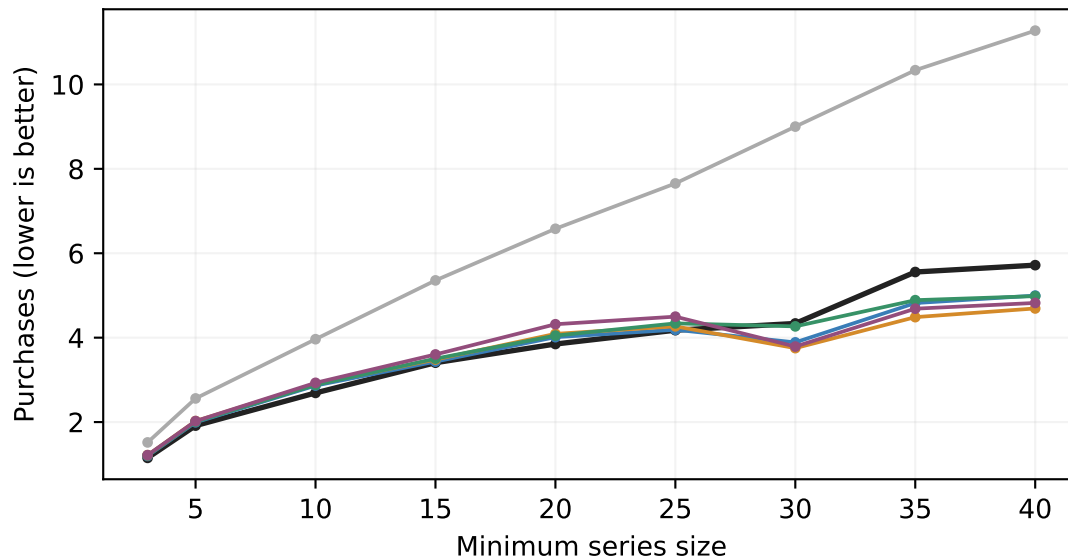
Purchases to any top-1



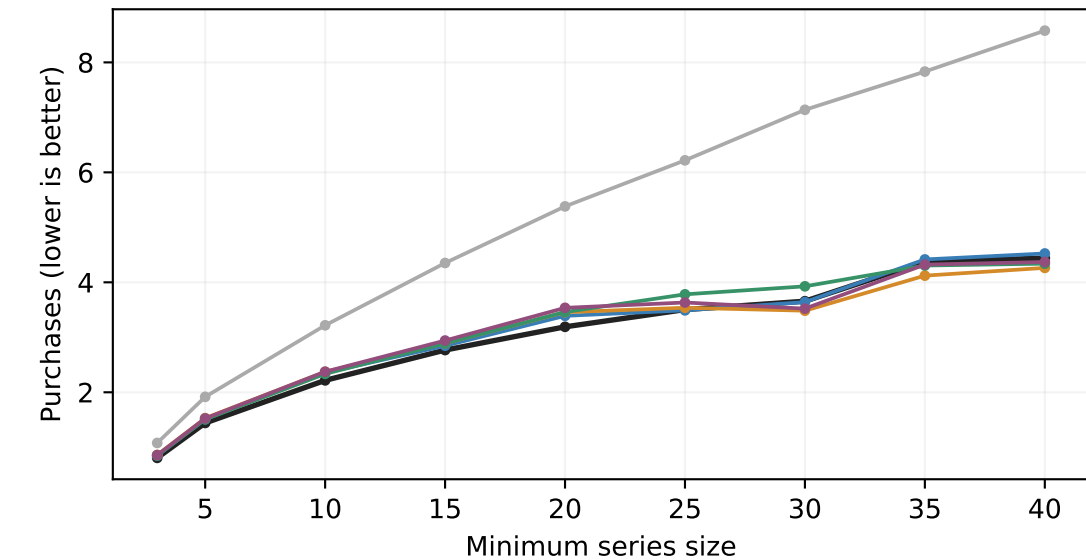
Purchases to any top-2



Purchases to any top-3



Purchases to any top-4

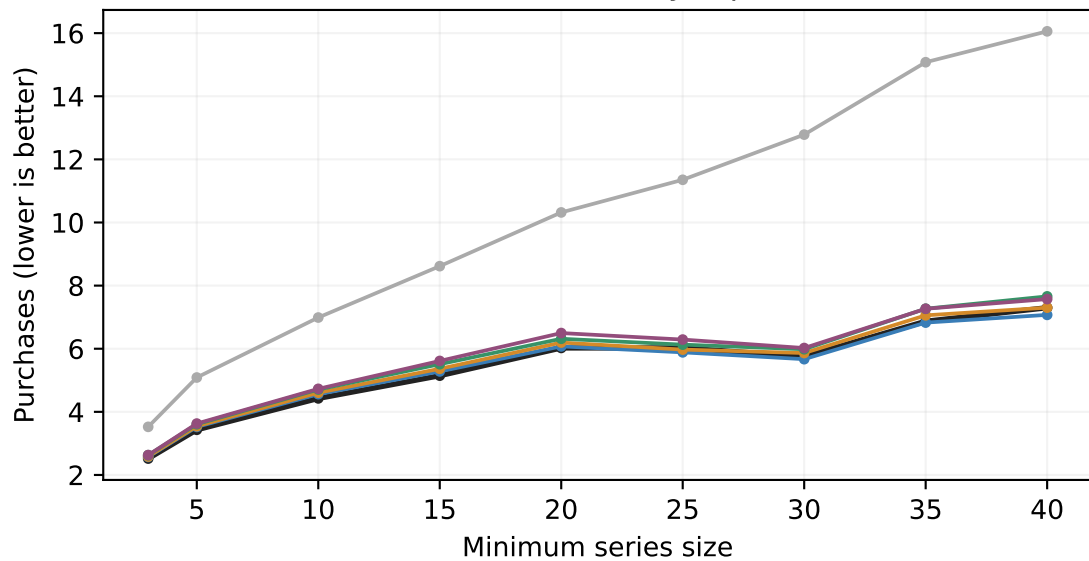


—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

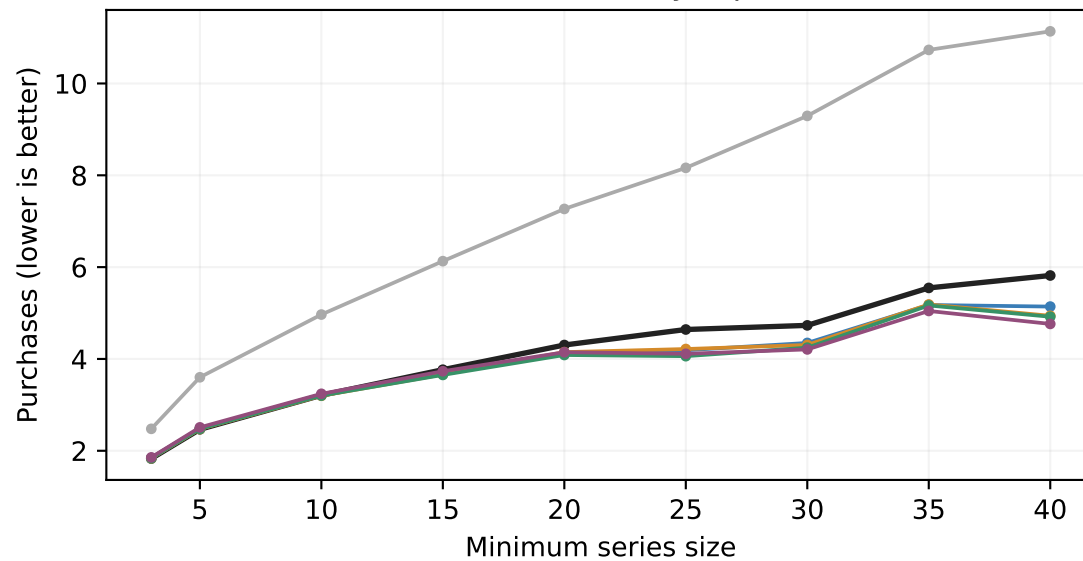
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Microsomal clearance

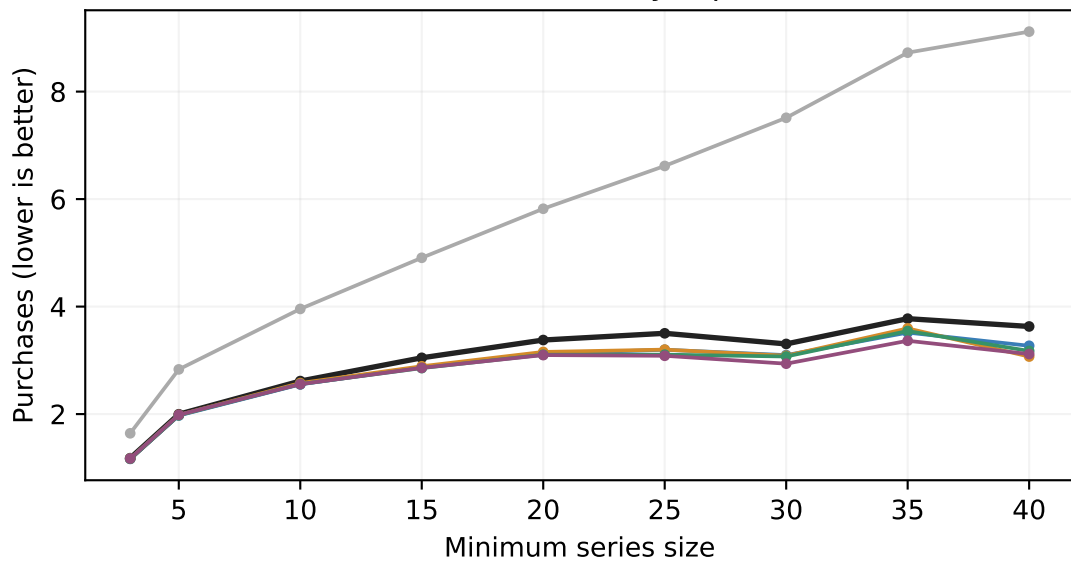
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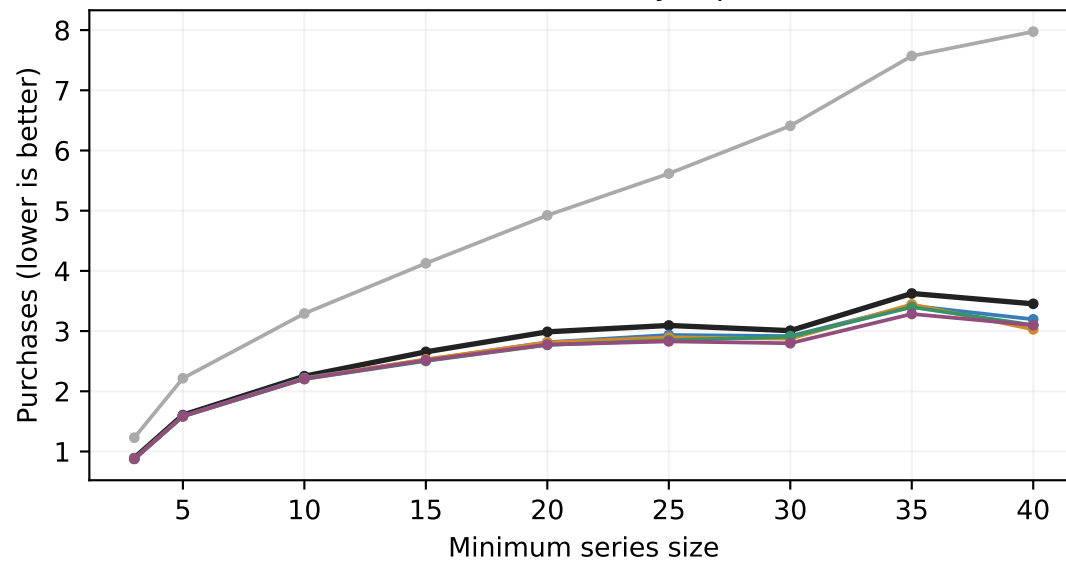
Purchases to any top-2



Purchases to any top-3



Purchases to any top-4

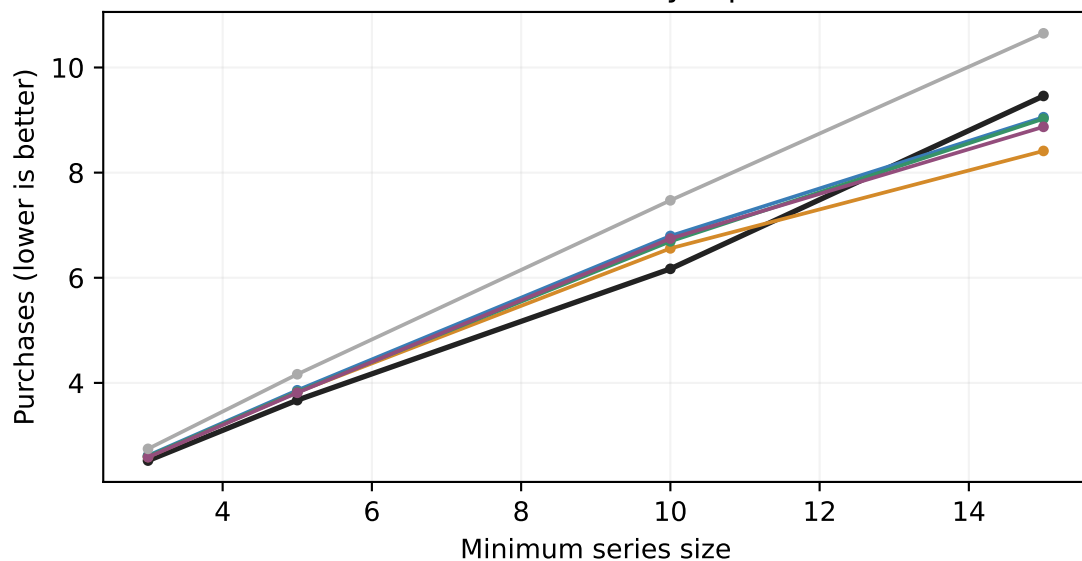


—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

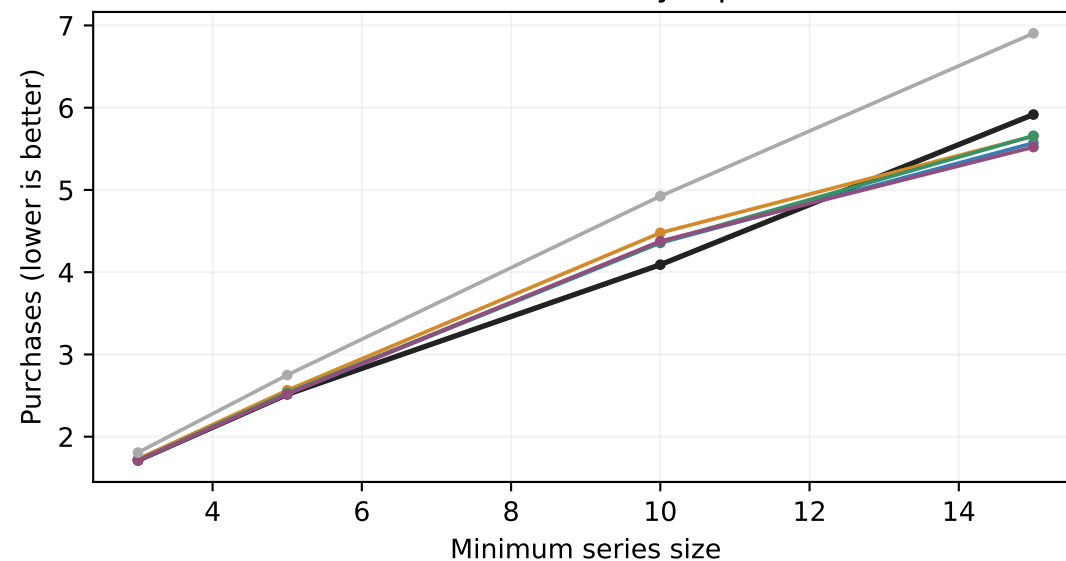
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In vivo clearance

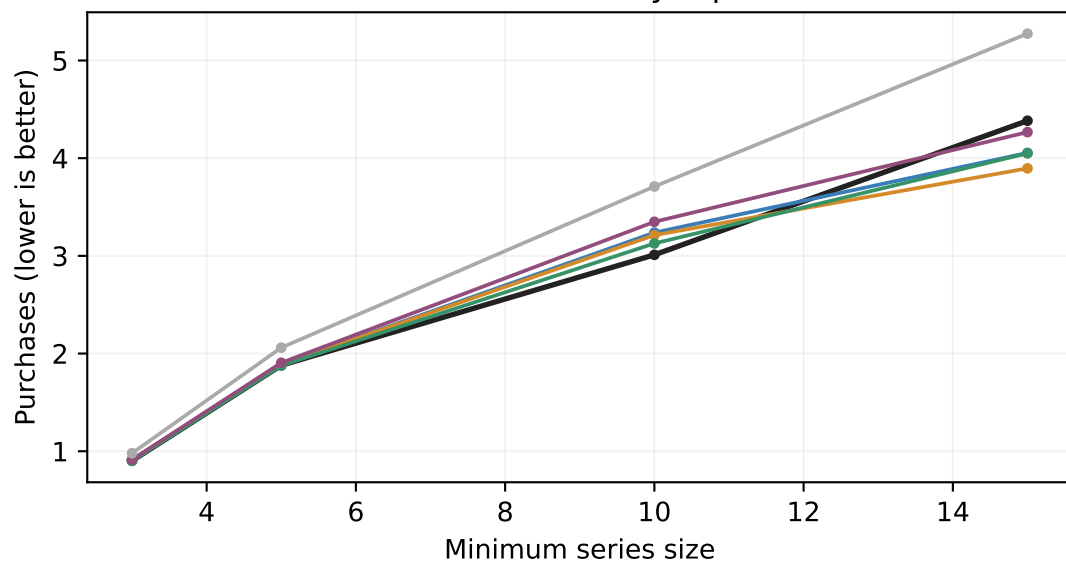
Purchases to any top-1



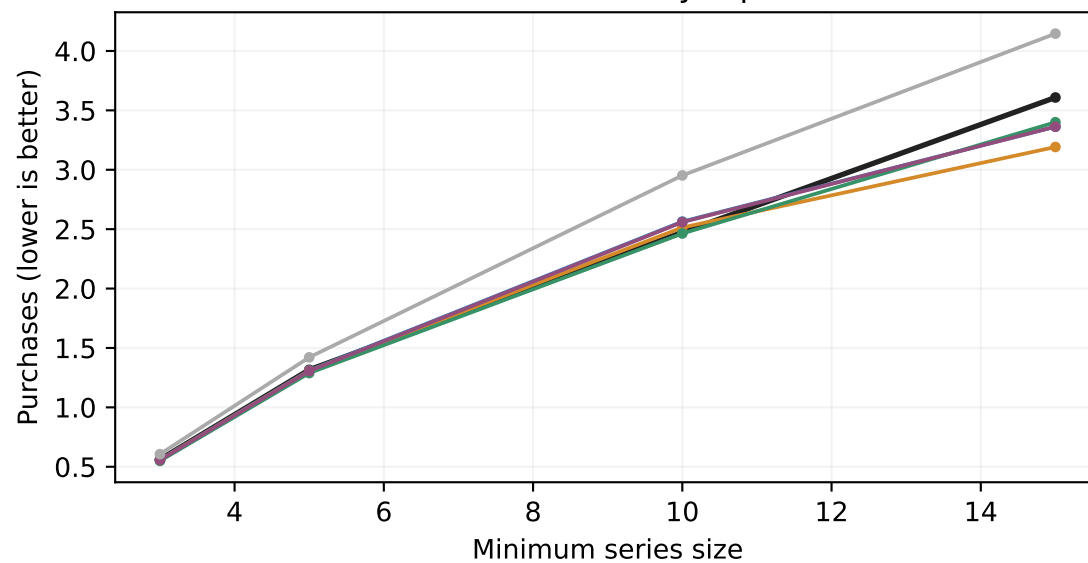
Purchases to any top-2



Purchases to any top-3



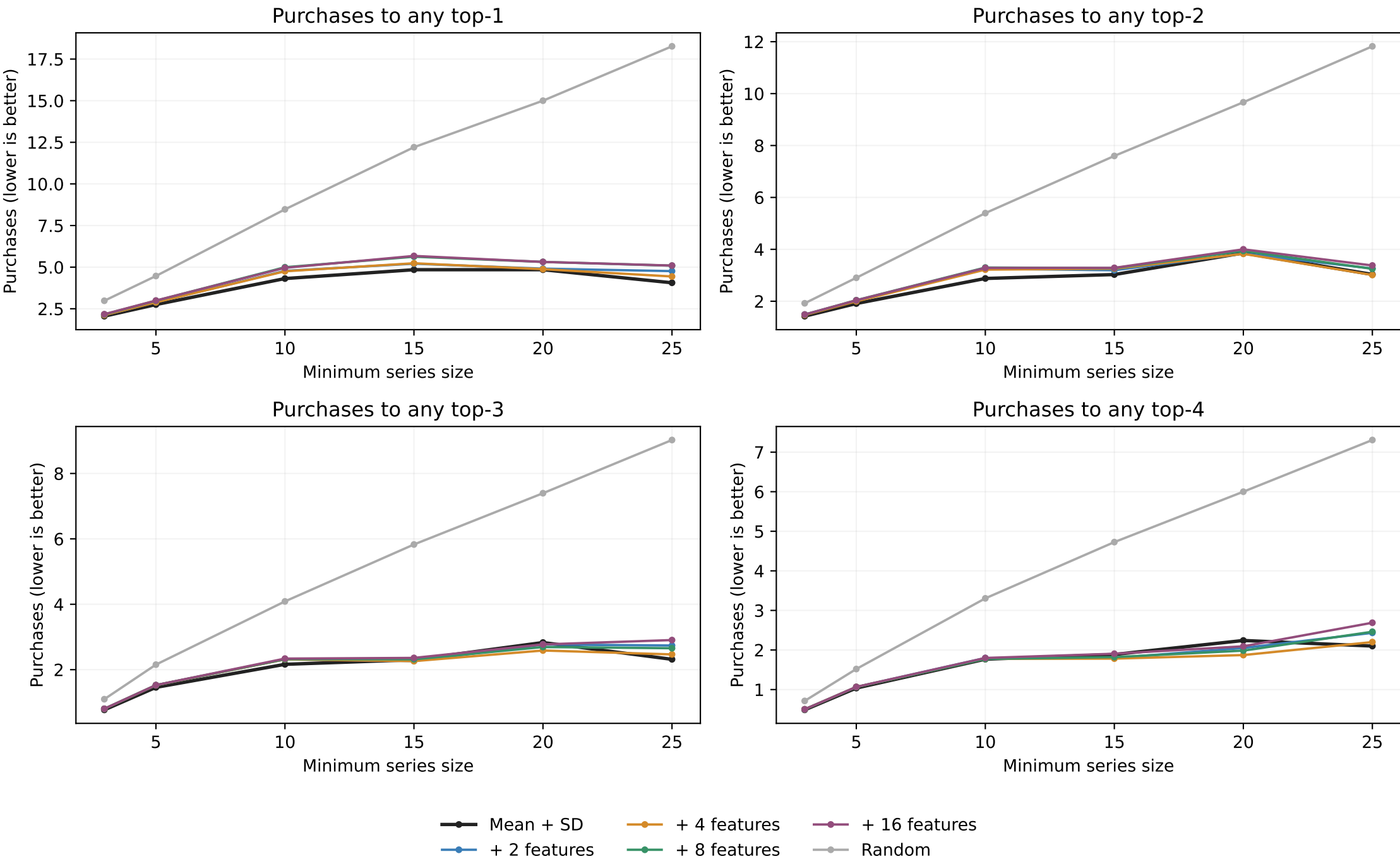
Purchases to any top-4



—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

Series at cutoffs 3/15/25/40: 537/20/5/1. High cutoffs are exploratory; endpoint averages receive equal weight.

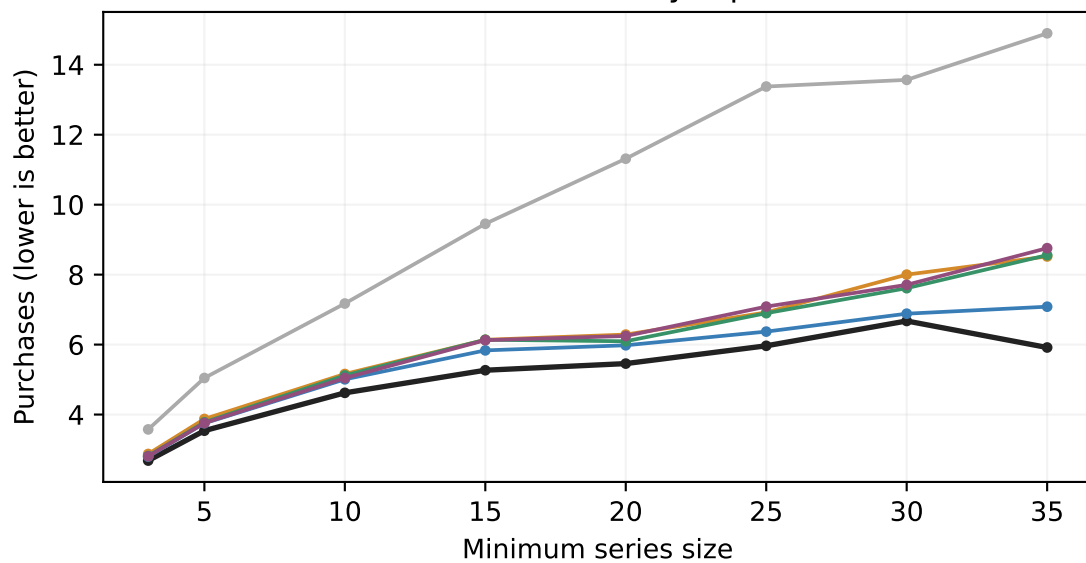
Protein binding



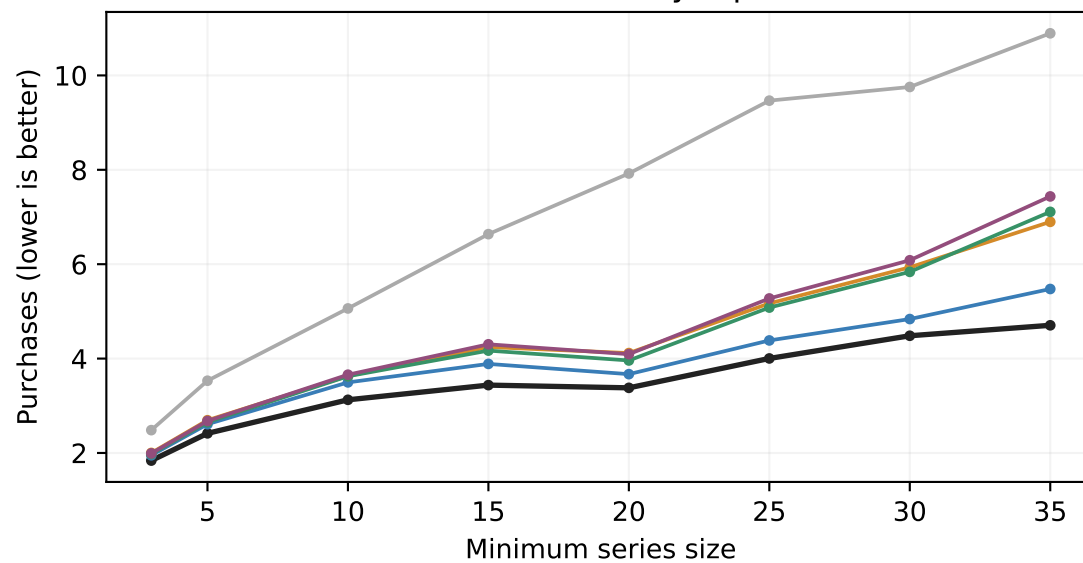
Series at cutoffs 3/15/25/40: 880/44/15/2. High cutoffs are exploratory; endpoint averages receive equal weight.

Cellular clearance

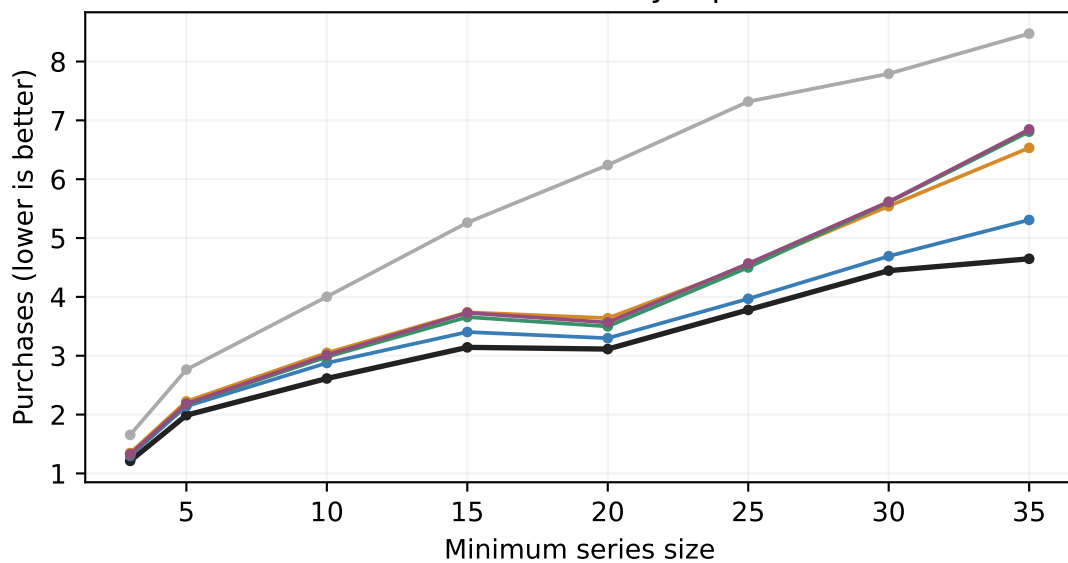
Purchases to any top-1



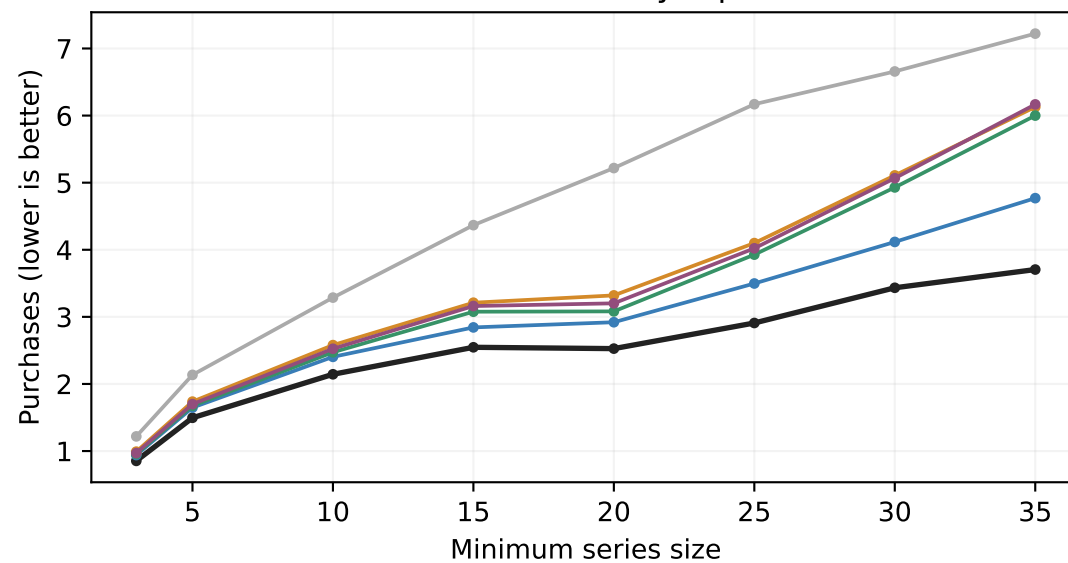
Purchases to any top-2



Purchases to any top-3



Purchases to any top-4

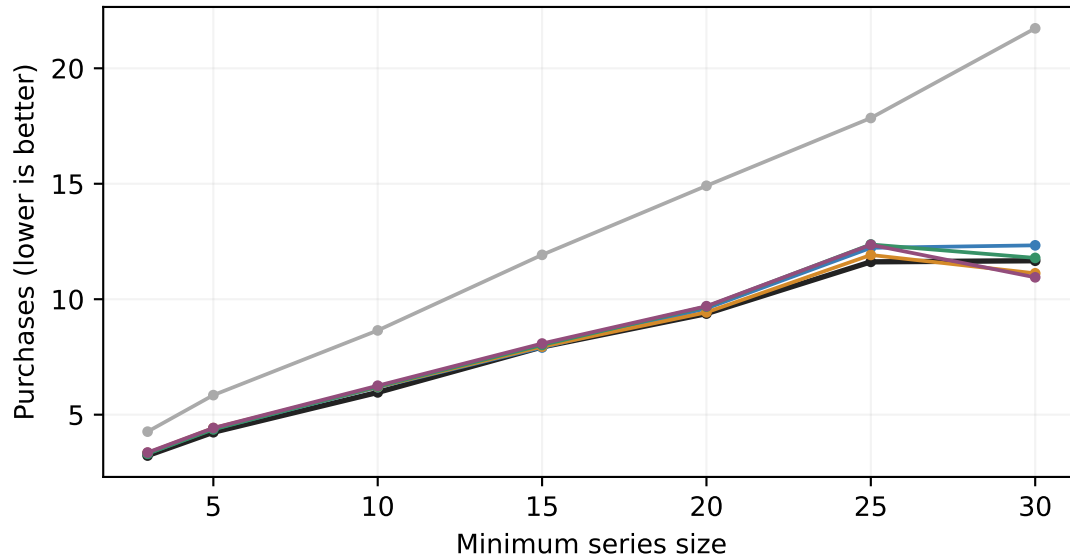


—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

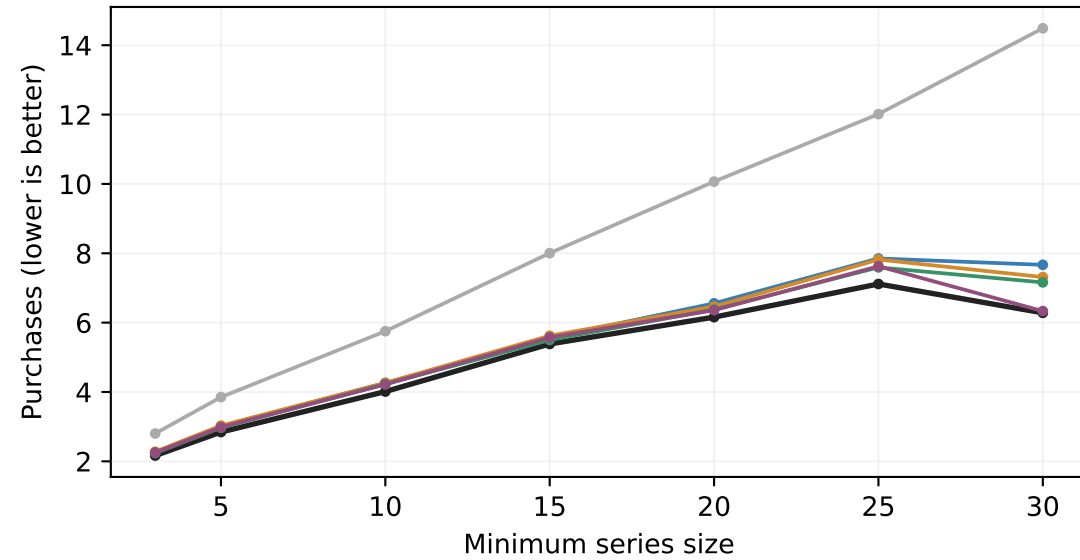
Series at cutoffs 3/15/25/40: 524/79/31/9. High cutoffs are exploratory; endpoint averages receive equal weight.

Permeability

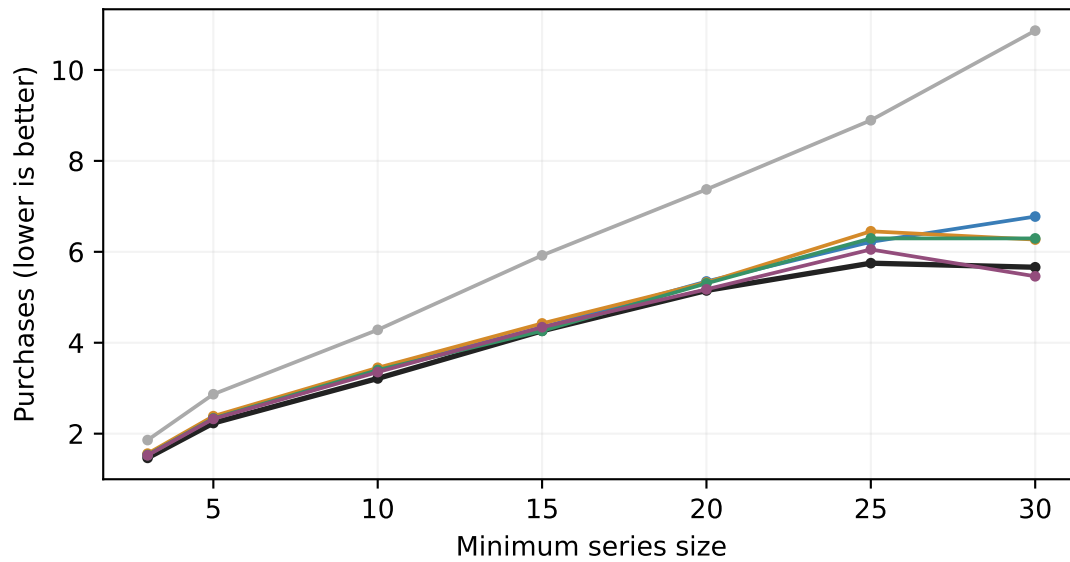
Purchases to any top-1



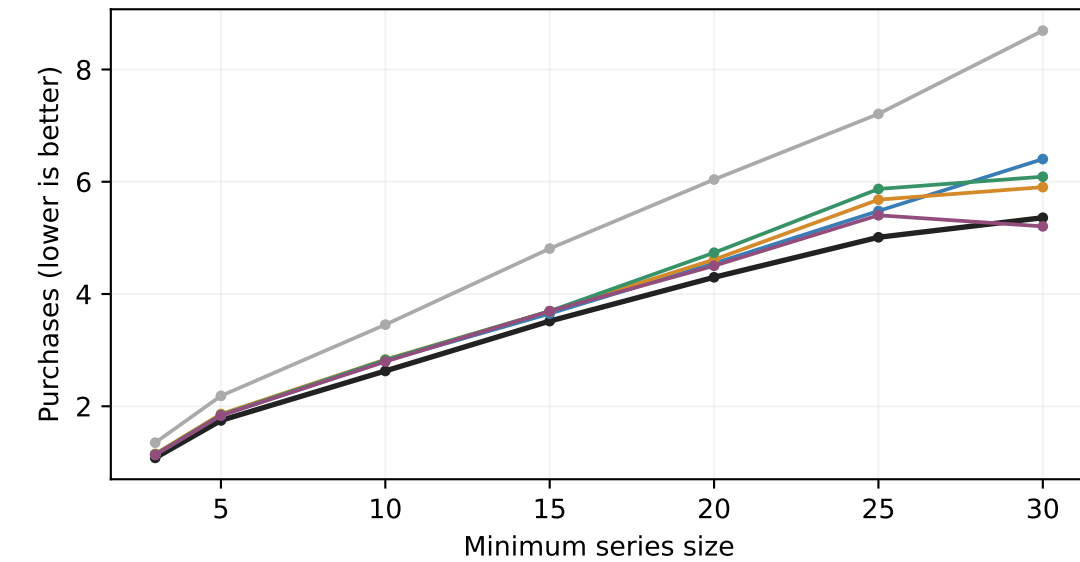
Purchases to any top-2



Purchases to any top-3



Purchases to any top-4

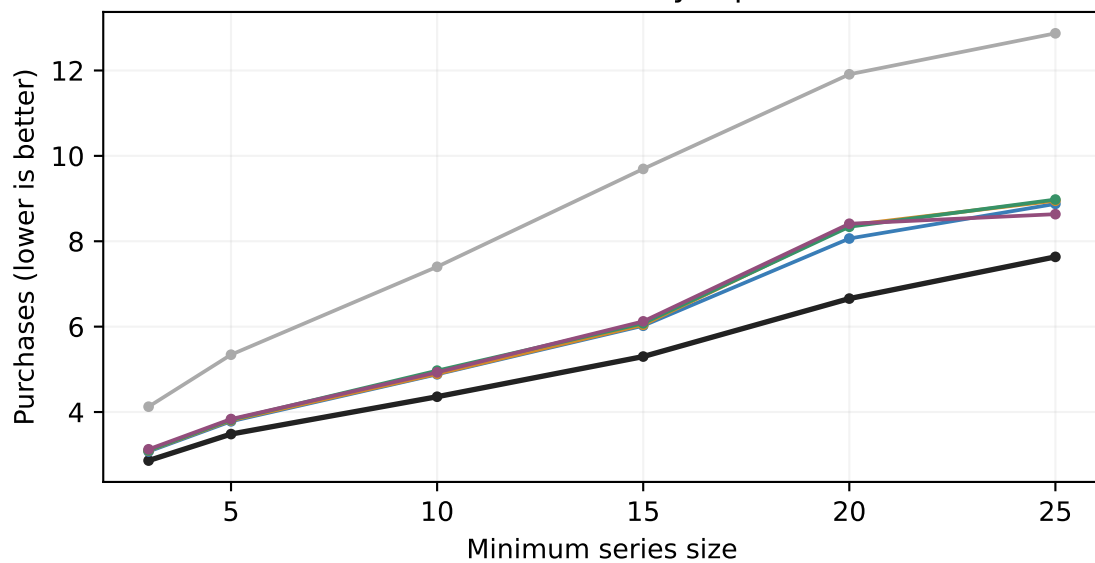


—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

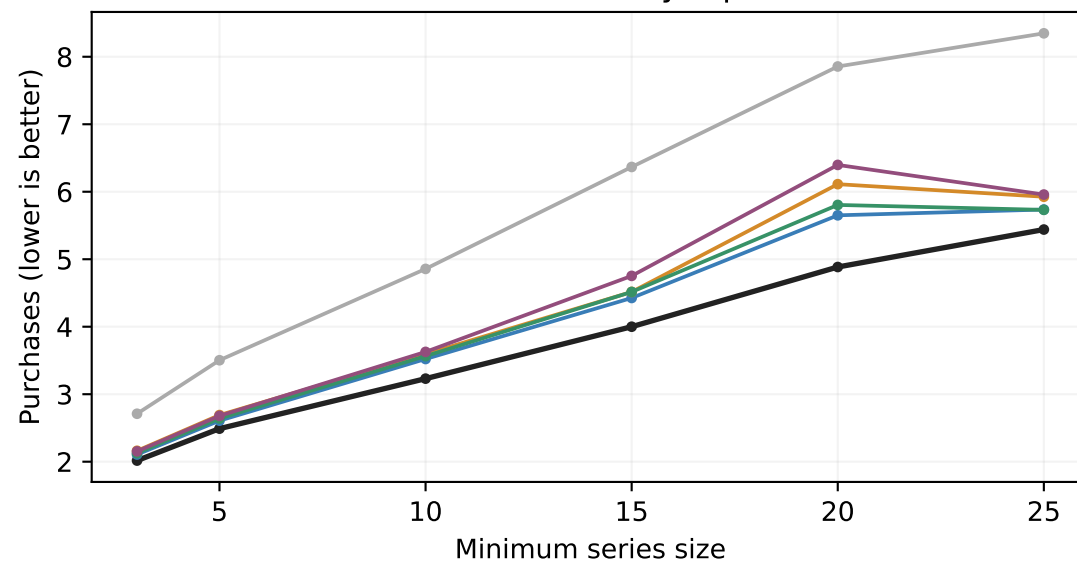
Series at cutoffs 3/15/25/40: 516/72/24/7. High cutoffs are exploratory; endpoint averages receive equal weight.

Efflux ratio

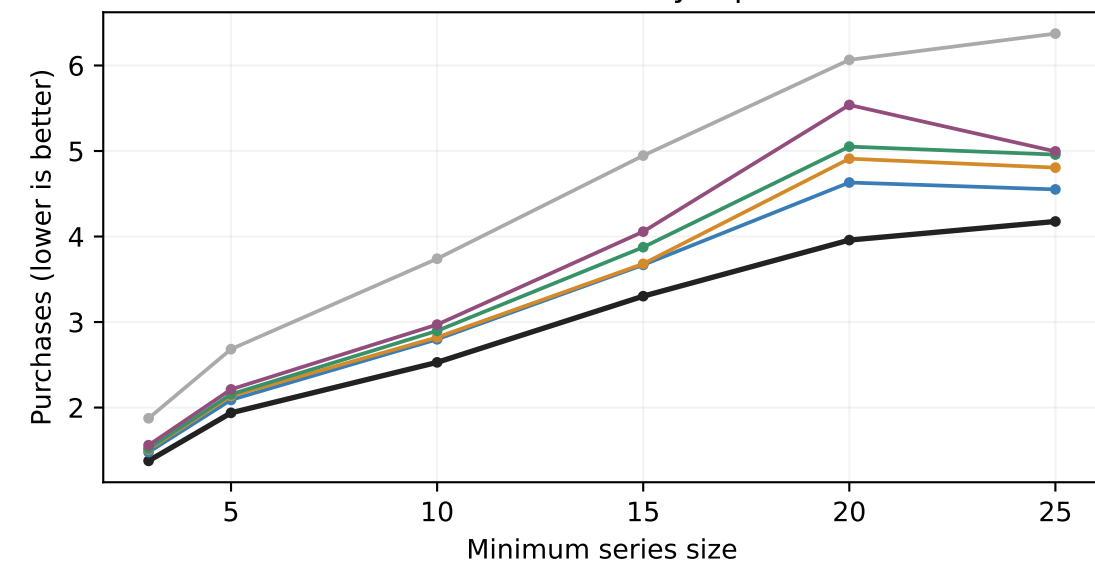
Purchases to any top-1



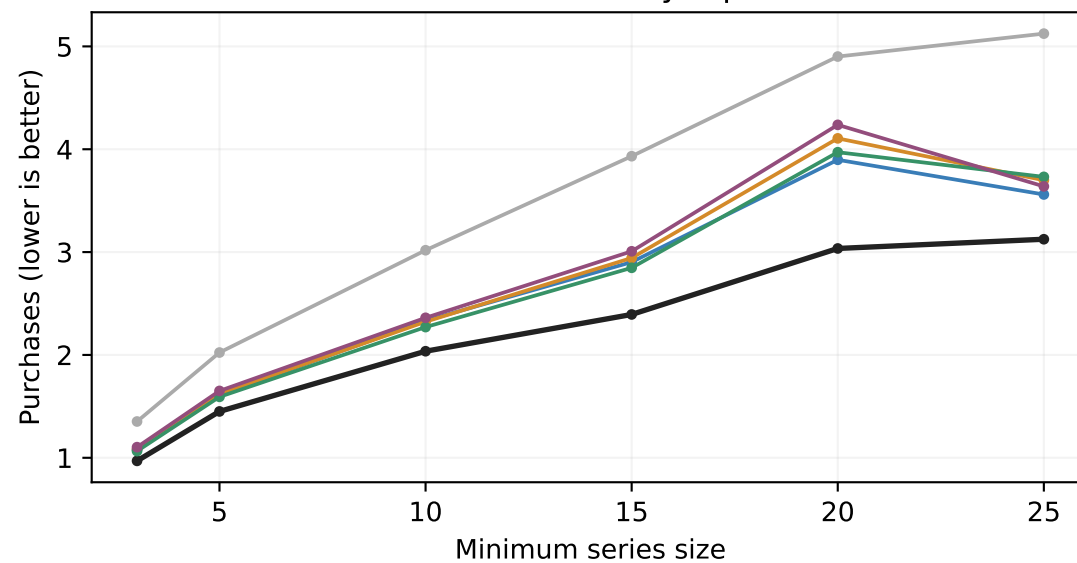
Purchases to any top-2



Purchases to any top-3



Purchases to any top-4

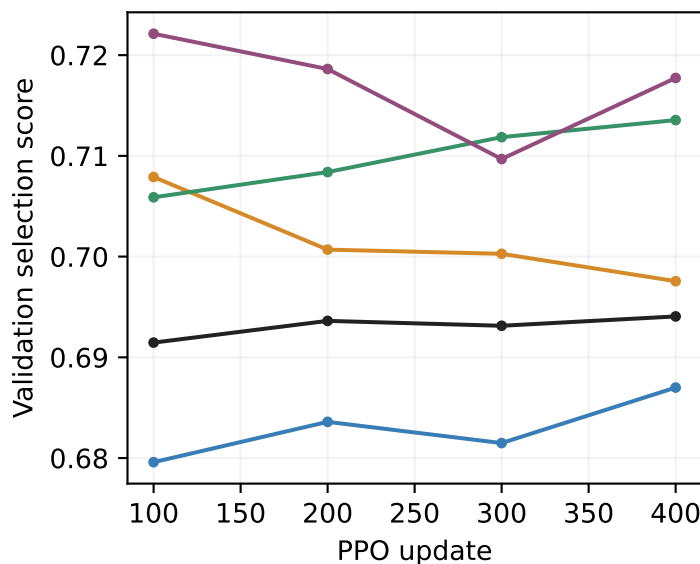


—●— Mean + SD —●— + 4 features —●— + 16 features
—●— + 2 features —●— + 8 features —●— Random

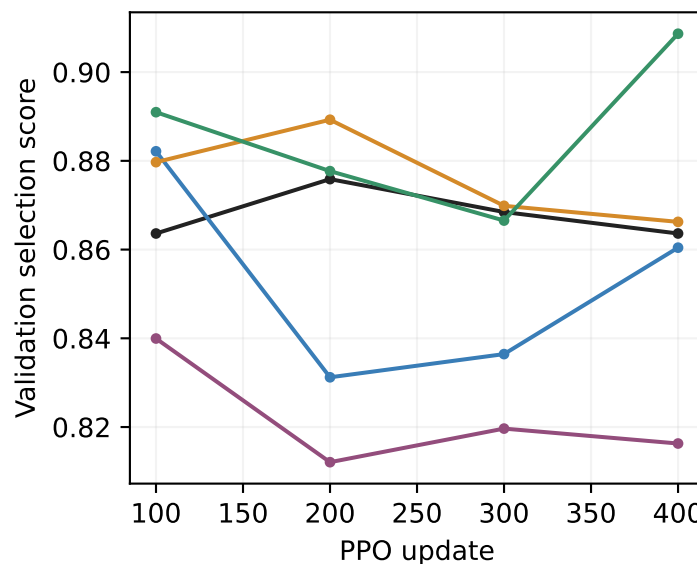
Series at cutoffs 3/15/25/40: 341/54/18/1. High cutoffs are exploratory; endpoint averages receive equal weight.

Validation learning curves (lower is better)

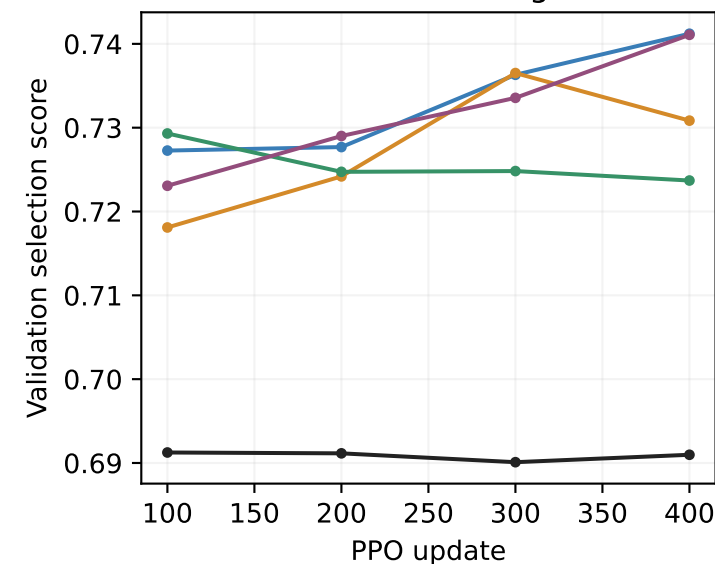
Microsomal clearance



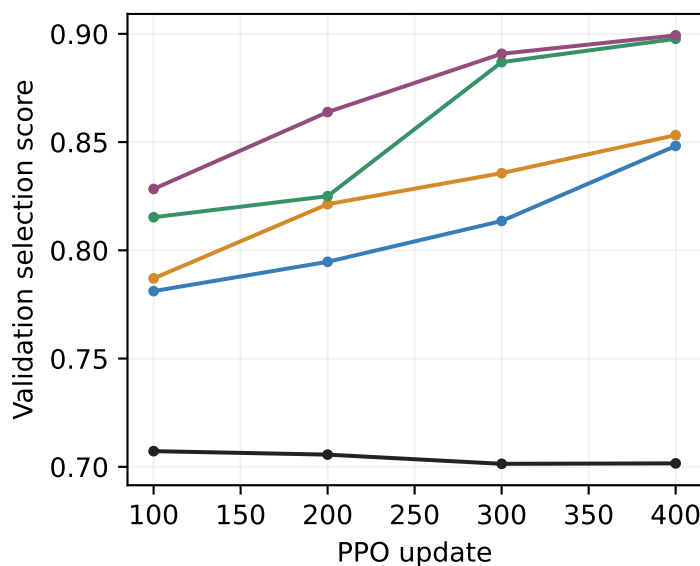
In vivo clearance



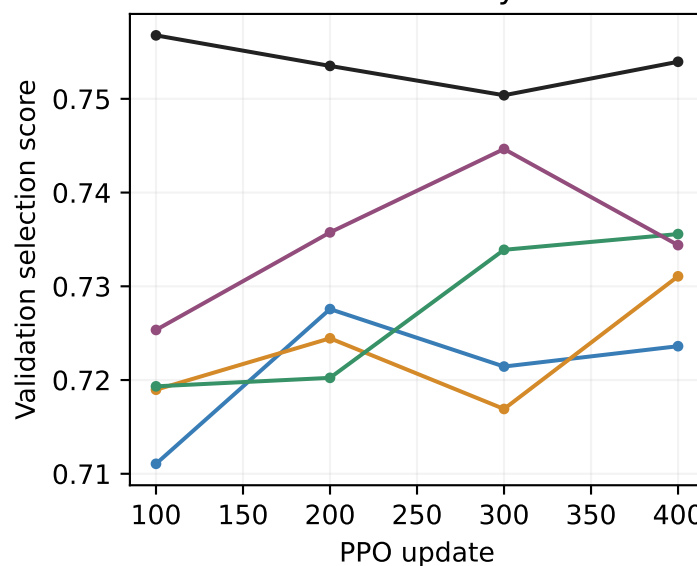
Protein binding



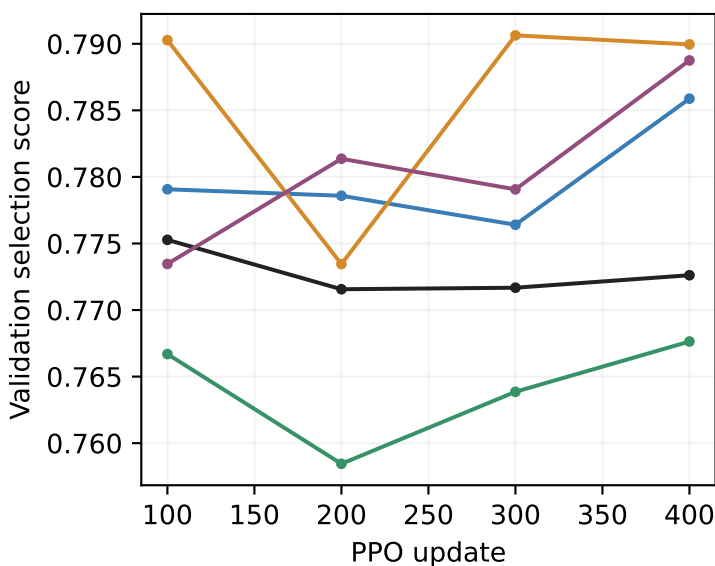
Cellular clearance



Permeability



Efflux ratio



—●— Mean + SD —●— + 2 features —●— + 4 features —●— + 8 features —●— + 16 features

Selection score: top-1 purchases divided by exact random expectation; half all pools, half large pools. Averaged across folds and seeds.

Architecture and training

Frozen prediction backbone: average pairwise delta mean + average-reference kernel; PCA32 MiniMol coordinates, width64,24 residual updates and conditional kernel variance. Three members fitted freshly on predictor-only training groups. Gaussian likelihood for two clearance tasks; Student-t for the others. Ensemble SD includes component variance and between-member mean disagreement.

Extra features: for a query, average its64-dimensional neural pair embedding across measured references separately for each predictor member. Concatenate the three summaries to192 dimensions; LayerNorm -> Linear(d) -> tanh. This avoids averaging unrelated latent coordinate bases across ensemble members.

Actor: [mean, SD, optional d features] ->64 tanh ->64 tanh -> one acquisition score. The same pointwise network is applied to every remaining candidate. No attention or candidate-set pooling enters the actor. PPO samples from softmax scores; deployment takes the largest score.

PPO: 400 updates,8 complete episodes/update, three optimization epochs, learning rate0.0003, clip0.2, entropy0.01. Reward -1 per purchase to the first tied optimum, without discounting. Initial hit is sampled uniformly from the worst half. Fresh actor and projection initialization, no EI imitation.

The critic is separate, identical across arms, and discarded at inference. It sees summary statistics of candidate mean/SD and remaining candidate count; it does not receive the extra embeddings.

Evaluation and limitations

Five outer folds keep linked molecules, documents and series together. Each outer-training set is divided approximately equally into predictor-training and policy-training linked groups. PCA uses outer-training structures only. Outcome scales use predictor-training outcomes only. Validation selects checkpoints; test is never used to select models.

Evaluate four common starting hits per series, then average the three independent policy seeds. Initial hits are free. Top-k means reaching any compound whose observed property is at least as good as the kth order statistic, including ties. If the initial hit already qualifies, its count is zero.

Random expectation is exact: when the initial hit is not a success, $n/(m+1)$, where n is total pool size and m is the number of remaining tied qualifying compounds. Lower acquisition count is better.

Confidence intervals use 3000 linked-group bootstrap resamples, preserving cross-endpoint group links. Intervals average over the three fixed policy seeds and do not include fresh backbone retraining uncertainty. Multiple feature dimensions and endpoints make the intervals exploratory.

The acquisition network hidden widths and PPO budget are held fixed. Extra features add input and projection parameters. This comparison does not establish the best possible feature dimension or training configuration. Prediction metrics do not change at identical measured states.

Passed audit: 277500 trajectories and 1110000 metric rows independently checked, including shared starts, stopping, ties, no repeated acquisitions, frozen checkpoint hashes and nonzero learned projection updates.